

# Experimental Design - Exam 1 (Take-home)

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## Problem 1

In a study of starting salaries for assistant professors, five male assistant professors at each of three types of institutions were randomly polled and their starting salaries (in \$1000) were recorded. The summary statistics are:

|          | Public | Private | Church-affiliated |
|----------|--------|---------|-------------------|
| Mean     | 53.4   | 70.8    | 56.1              |
| Variance | 72.8   | 48.2    | 60.3              |

There are  $n_i = 5$  observations per group ( $a = 3$ ,  $N = 15$ ).

### (a) Type of experimental design and model

This is a single-factor completely randomized design (CRD) with one factor “institution type” at three levels. A fixed-effects one-way ANOVA model is

$$Y_{ij} = \mu + \tau_i + \varepsilon_{ij}, \quad i = 1, 2, 3, \quad j = 1, \dots, 5,$$

where  $\mu$  is the overall mean,  $\tau_i$  is the effect of institution type  $i$  with the usual constraint  $\sum_{i=1}^3 \tau_i = 0$ , and  $\varepsilon_{ij} \sim \text{NID}(0, \sigma^2)$ .

### (b) Test for differences in average starting salaries

We test

$$H_0 : \mu_1 = \mu_2 = \mu_3 \quad \text{vs} \quad H_a : \text{at least one } \mu_i \text{ differs.}$$

The grand mean is

$$\bar{y}_{..} = \frac{53.4 + 70.8 + 56.1}{3} = 60.1.$$

The treatment sum of squares is

$$SS_{\text{Trt}} = n \sum_{i=1}^3 (\bar{y}_i - \bar{y}_{..})^2 = 5[(53.4 - 60.1)^2 + (70.8 - 60.1)^2 + (56.1 - 60.1)^2] = 876.9.$$

The error sum of squares is

$$SS_E = \sum_{i=1}^3 (n_i - 1)s_i^2 = 4(72.8 + 48.2 + 60.3) = 725.2,$$

so the total sum of squares is

$$SS_T = SS_{\text{Trt}} + SS_E = 876.9 + 725.2 = 1602.1.$$

Degrees of freedom and mean squares:

$$df_{\text{Trt}} = a - 1 = 2, \quad df_E = N - a = 12, \quad df_T = N - 1 = 14,$$

$$MS_{\text{Trt}} = \frac{SS_{\text{Trt}}}{2} = 438.45, \quad MS_E = \frac{SS_E}{12} = 60.4333.$$

The ANOVA  $F$ -statistic is

$$F = \frac{MS_{\text{Trt}}}{MS_E} = \frac{438.45}{60.4333} \approx 7.26,$$

which is compared to an  $F$  distribution with  $(2, 12)$  degrees of freedom. The corresponding  $p$ -value from R is

$$p = 0.0086.$$

Since  $p < 0.05$ , we reject  $H_0$  and conclude that there are significant differences in average starting salaries among the three institution types.

### (c) Estimate of the standard deviation of the error

The error mean square  $MS_E$  is the usual unbiased estimator of  $\sigma^2$ . Hence

$$\hat{\sigma}^2 = MS_E = 60.4333, \quad \hat{\sigma} = \sqrt{MS_E} \approx 7.77,$$

so the estimated error standard deviation is about \$7,770.

### (d) Pairwise comparisons using Tukey HSD and Scheffé

With a balanced design and common replication  $n = 5$ , the standard error for pairwise differences of treatment means is

$$SE(\bar{y}_i - \bar{y}_j) = \sqrt{\frac{2MS_E}{n}} = \sqrt{\frac{2 \times 60.4333}{5}} = \sqrt{24.1733} \approx 4.92.$$

The pairwise sample mean differences (treatment  $j$  minus treatment  $i$ ) are

$$\bar{y}_{\text{Priv}} - \bar{y}_{\text{Pub}} = 17.4, \quad \bar{y}_{\text{Ch}} - \bar{y}_{\text{Pub}} = 2.7, \quad \bar{y}_{\text{Ch}} - \bar{y}_{\text{Priv}} = -14.7.$$

*Tukey HSD.* For  $\alpha = 0.05$ ,  $a = 3$ , and  $df_E = 12$ , the studentized range critical value from R is

$$q_{0.05, 3, 12} = 3.7729,$$

so

$$HSD = q_{0.05, 3, 12} \sqrt{\frac{MS_E}{n}} = 3.7729 \sqrt{\frac{60.4333}{5}} \approx 3.7729 \times 3.48 \approx 13.1.$$

A pair  $(i, j)$  is significantly different if  $|\bar{y}_i - \bar{y}_j| > HSD$ .

- Private vs. Public:  $|\bar{y}_{\text{Priv}} - \bar{y}_{\text{Pub}}| = 17.4 > 13.1 \Rightarrow \text{significant}.$

- Church vs. Public:  $|\bar{y}_{\text{Ch}} - \bar{y}_{\text{Pub}}| = 2.7 < 13.1 \Rightarrow$  not significant.
- Church vs. Private:  $|\bar{y}_{\text{Ch}} - \bar{y}_{\text{Priv}}| = 14.7 > 13.1 \Rightarrow$  significant.

Thus, under Tukey's method, Private institutions have significantly higher starting salaries than both Public and Church-affiliated institutions, while Public and Church-affiliated institutions do not differ significantly.

*Scheffé method.* For a one-way ANOVA with  $a = 3$  groups, Scheffé's critical factor at  $\alpha = 0.05$  is

$$k_S = (a - 1)F_{0.05, a-1, df_E} = 2F_{0.05, 2, 12}.$$

From R,

$$F_{0.05, 2, 12} = 3.8853, \quad k_S = 2 \times 3.8853 \approx 7.77.$$

For a pairwise contrast  $C = \mu_i - \mu_j$ , the estimated contrast is  $\hat{C} = \bar{y}_i - \bar{y}_j$  with

$$SE(\hat{C}) = SE(\bar{y}_i - \bar{y}_j) \approx 4.92,$$

and Scheffé's test rejects  $H_0 : C = 0$  if

$$\frac{\hat{C}^2}{SE(\hat{C})^2} > k_S.$$

Using the R output,

$$\begin{aligned} T_{\text{Priv-Pub}}^2 &= 12.52, \\ T_{\text{Ch-Pub}}^2 &= 0.30, \\ T_{\text{Ch-Priv}}^2 &= 8.94, \end{aligned}$$

so

$$T_{\text{Priv-Pub}}^2 > 7.77, \quad T_{\text{Ch-Priv}}^2 > 7.77, \quad T_{\text{Ch-Pub}}^2 < 7.77.$$

Therefore, Scheffé's method also finds that Private vs. Public and Private vs. Church-affiliated are significantly different, while Public vs. Church-affiliated is not. In this example, Tukey and Scheffé give the same pairwise conclusions.

### (e) Kruskal–Wallis test

A Kruskal–Wallis test with  $a = 3$  groups has  $df = a - 1 = 2$  and, under  $H_0$ , the test statistic is approximately  $\chi_2^2$ -distributed. We are told that the Kruskal–Wallis test statistic is  $H = 8.09$ . The 5% upper critical value for  $\chi_2^2$  is about 5.99, and since  $8.09 > 5.99$ , the Kruskal–Wallis test also rejects  $H_0$  at the 5% level. Thus the nonparametric analysis confirms the ANOVA result: there is evidence of differences in starting salaries among the three institution types, so our conclusion in part (b) does not change.

### (f) Possible improved model

The current model treats only institution type as a factor and ignores other important sources of variability. An improved model might:

- include additional factors or covariates such as academic discipline, geographic region, or research vs. teaching focus, which are known to affect starting salaries;

- use blocking on departments, regions, or year of hire to control for nuisance variation via a randomized complete block design;
- consider transformations (e.g., modeling log-salary) if residual plots indicate skewness or nonconstant variance across institution types.

Such extensions would likely reduce unexplained variability and provide a more realistic assessment of the effect of institution type on starting salaries.

### Problem 2

Three groups (#1, #2, #3) of students were subjected to different teaching techniques and tested at the end of the course. There are  $n_i = 6$  students in each group, and the average scores are

$$\bar{y}_1 = 72.3, \quad \bar{y}_2 = 79.5, \quad \bar{y}_3 = 65.8.$$

The overall variance of all 18 scores is 84.03, and we are told that the error variance estimate from the ANOVA model is

$$MS_E = 57.84.$$

#### (a) Complete ANOVA table

The grand mean is

$$\bar{y}_{..} = \frac{72.3 + 79.5 + 65.8}{3} = 72.5333.$$

The treatment sum of squares is

$$SS_{\text{Trt}} = n \sum_{i=1}^3 (\bar{y}_i - \bar{y}_{..})^2 = 6[(72.3 - 72.5333)^2 + (79.5 - 72.5333)^2 + (65.8 - 72.5333)^2] = 563.56.$$

We are given  $MS_E = 57.84$ . The error degrees of freedom are

$$df_E = N - a = 18 - 3 = 15,$$

so

$$SS_E = MS_E \times df_E = 57.84 \times 15 = 867.6.$$

Thus the total sum of squares is

$$SS_T = SS_{\text{Trt}} + SS_E = 563.56 + 867.6 = 1431.16.$$

Degrees of freedom and mean squares:

$$df_{\text{Trt}} = a - 1 = 2, \quad df_E = 15, \quad df_T = N - 1 = 17,$$

$$MS_{\text{Trt}} = \frac{SS_{\text{Trt}}}{2} = 281.78, \quad MS_E = 57.84.$$

The ANOVA  $F$ -statistic is

$$F = \frac{MS_{\text{Trt}}}{MS_E} = \frac{281.78}{57.84} \approx 4.87.$$

The ANOVA table is therefore

| Source    | df | SS      | MS     | <i>F</i> |
|-----------|----|---------|--------|----------|
| Treatment | 2  | 563.56  | 281.78 | 4.87     |
| Error     | 15 | 867.60  | 57.84  |          |
| Total     | 17 | 1431.16 |        |          |

**(b) Conclusion about differences in teaching techniques**

We test

$$H_0 : \mu_1 = \mu_2 = \mu_3 \quad \text{vs} \quad H_a : \text{at least one } \mu_i \text{ differs.}$$

From the ANOVA table,

$$F = 4.8717, \quad df_1 = 2, \quad df_2 = 15.$$

The corresponding *p*-value from R is

$$p = 0.0234.$$

Since  $p < 0.05$ , we reject  $H_0$  at the 5% level and conclude that there are significant differences in mean test scores among the three teaching techniques.

**(c) Comparison of treatments to the control (group 2)**

Suppose group 2 is the control and groups 1 and 3 are treatments. The differences in sample means relative to the control are

$$\bar{y}_1 - \bar{y}_2 = 72.3 - 79.5 = -7.2, \quad \bar{y}_3 - \bar{y}_2 = 65.8 - 79.5 = -13.7.$$

The standard error for the difference between a treatment mean and the control mean (with equal group sizes  $n = 6$ ) is

$$SE_{\text{diff}} = \sqrt{\frac{2MS_E}{n}} = \sqrt{\frac{2 \times 57.84}{6}} = 4.3909.$$

Thus the *t*-statistics (with  $df = 15$ ) are

$$t_{1-2} = \frac{-7.2}{4.3909} = -1.64, \quad t_{3-2} = \frac{-13.7}{4.3909} = -3.12.$$

At the usual  $\alpha = 0.05$  level for a two-sided test, the comparison of group 3 to group 2 is significant (group 3 has a significantly lower mean score than the control), while the comparison of group 1 to group 2 is not statistically significant, even though group 1's mean is somewhat lower than the control.

**(d) Average treatment effect vs control with Bonferroni correction**

We now test whether the *average* of the two treatment groups differs from the control. Consider the contrast

$$C = \frac{1}{2}\mu_1 + \frac{1}{2}\mu_3 - \mu_2.$$

Its estimate is

$$\hat{C} = \frac{1}{2}(72.3 + 65.8) - 79.5 = \frac{1}{2}(138.1) - 79.5 = 69.05 - 79.5 = -10.45.$$

For equal group sizes  $n = 6$ , the variance of  $\hat{C}$  is

$$\text{Var}(\hat{C}) = MS_E \left( \frac{0.5^2}{n} + \frac{0.5^2}{n} + \frac{1^2}{n} \right) = MS_E \cdot \frac{1.5}{n},$$

so

$$SE_C = \sqrt{\frac{1.5 MS_E}{n}} = \sqrt{\frac{1.5 \times 57.84}{6}} = 3.8026.$$

The corresponding  $t$ -statistic (with  $df = 15$ ) is

$$t = \frac{\hat{C}}{SE_C} = \frac{-10.45}{3.8026} = -2.75.$$

We are asked to test this contrast using a Bonferroni correction assuming a total of  $K = 10$  post-hoc tests. With overall  $\alpha = 0.05$ , the Bonferroni-adjusted per-test level is

$$\alpha^* = \frac{\alpha}{K} = \frac{0.05}{10} = 0.005.$$

For a two-sided test, we use  $\alpha^*/2 = 0.0025$  in each tail. The critical  $t$  value from R is

$$t_{1-\alpha^*/2, 15} = t_{0.9975, 15} = 3.286.$$

Since

$$|t| = 2.75 < 3.286,$$

we fail to reject  $H_0 : C = 0$  at the Bonferroni-adjusted level. Thus, when controlling for a family of 10 post-hoc tests, the average of the two treatment groups is not significantly different from the control group, even though the unadjusted  $t$ -statistic suggests a noticeable decrease in score.

### Problem 3 (DAE 4.17)

The response is gene expression measured for three leukemia treatments: MP only, MP + HDMTX, and MP + LDMTX, with  $n = 10$  observations per treatment ( $N = 30$ ). The data are in `Mid1Prob3.csv`.

#### (a) Test whether treatment means differ (raw data)

A one-way fixed-effects ANOVA model is

$$Y_{ij} = \mu + \tau_i + \varepsilon_{ij}, \quad i = 1, 2, 3, \quad j = 1, \dots, 10, \quad \varepsilon_{ij} \sim \text{NID}(0, \sigma^2),$$

where  $i$  indexes treatment.

Sample means and standard deviations by treatment are

| Treatment  | $n$ | Mean  | SD    |
|------------|-----|-------|-------|
| MP + HDMTX | 10  | 479.3 | 371.0 |
| MP + LDMTX | 10  | 165.0 | 126.0 |
| MP Only    | 10  | 238.0 | 308.0 |

From the ANOVA on the raw responses (R command `aov(Gene.Expression ~ Treatment)`),

| Source    | Df | SS        | MS      | $F$  |
|-----------|----|-----------|---------|------|
| Treatment | 2  | 538,442   | 269,221 | 3.25 |
| Error     | 27 | 2,236,974 | 82,851  |      |

with  $p$ -value  $p = 0.0544$ .

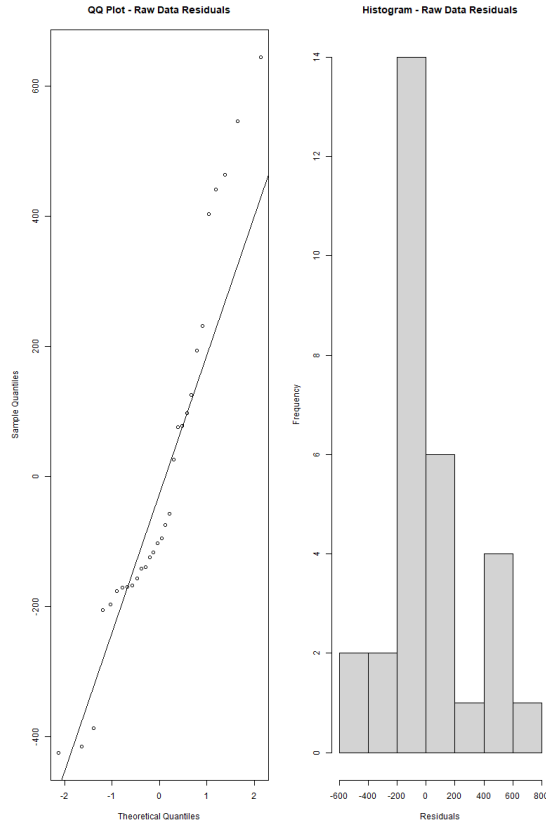
At  $\alpha = 0.05$ ,  $p$  is slightly larger than 0.05, so we *do not* reject  $H_0$  at the 5% level based strictly on this ANOVA. The evidence for a difference in mean gene expression across treatments is marginal but not conventionally significant.

### (b) Normality check for raw data

The residuals from the raw-data ANOVA show substantial deviation from normality. The Shapiro–Wilk test applied to the raw residuals gives

$$W = 0.9165, \quad p = 0.0218,$$

which is below 0.05, indicating a statistically significant departure from normality. The residual histogram and QQ plot exhibit pronounced right skew and a few large positive residuals, consistent with the presence of outliers or heavy right tails. Thus the normality assumption is questionable on the original scale.



**(c) Log transform and ANOVA on transformed data**

Because the raw data are positive and strongly right-skewed, a log transform is natural. Let

$$Z_{ij} = \log(Y_{ij}),$$

and fit the one-way ANOVA model

$$Z_{ij} = \mu^* + \tau_i^* + \varepsilon_{ij}^*.$$

On the log scale, the sample means and standard deviations are

| Treatment  | $n$ | Mean | SD   |
|------------|-----|------|------|
| MP + HDMTX | 10  | 5.73 | 1.13 |
| MP + LDMTX | 10  | 4.74 | 1.00 |
| MP Only    | 10  | 4.79 | 1.18 |

The ANOVA on the log-transformed responses (R command `aov(logGE ~ Treatment)`) yields

| Source    | Df | SS     | MS     | $F$  |
|-----------|----|--------|--------|------|
| Treatment | 2  | 6.297  | 3.1485 | 2.57 |
| Error     | 27 | 33.069 | 1.2248 |      |

with  $p = 0.0951$ .

Thus, on the log scale the evidence for treatment differences is weaker: at  $\alpha = 0.05$  we again fail to reject  $H_0$ ; the  $p$ -value is between 0.05 and 0.10, so one might describe this as only marginal evidence of differences in mean log gene expression.

**(d) Residual analysis for the transformed model**

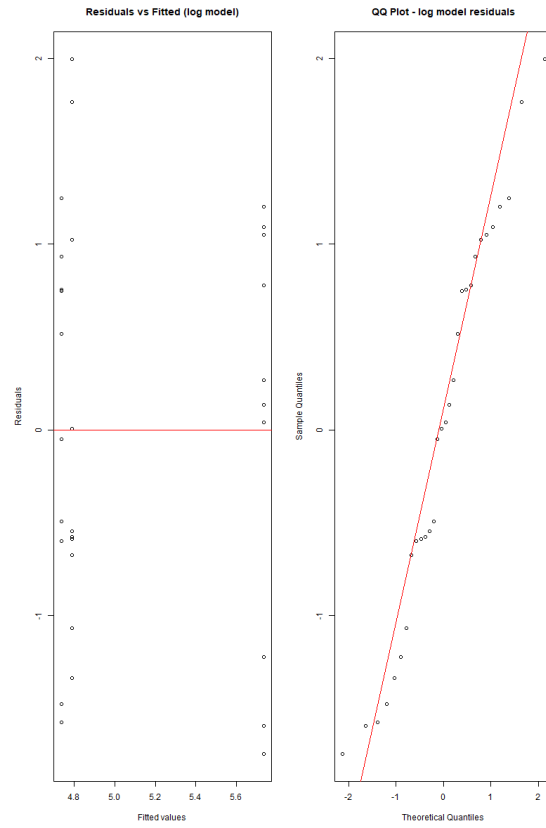
For the log-scale ANOVA, we examine the residuals. The residuals vs fitted plot for the log model shows a fairly symmetric cloud of points with no strong funnel shape; the variability appears roughly constant across fitted values. The QQ plot of the log-model residuals lies much closer to a straight line than for the raw data.

The Shapiro–Wilk test on the log-model residuals gives

$$W = 0.9565, \quad p = 0.2511,$$

which is not significant; we do not reject normality at the 5% level. This suggests that the log transform greatly improves the normality of the errors and makes the ANOVA assumptions more plausible.

In summary, the log transformation yields a model with much better residual behavior (approximately normal and homoscedastic), but even on this scale the treatment effect is not statistically significant at the  $\alpha = 0.05$  level.



## R Code

```
> #####
> # Problem 1
> #####
>
> # 1(a-d): One-way ANOVA from summary stats
> means <- c(Public = 53.4, Private = 70.8, Church = 56.1)
> vars   <- c(Public = 72.8, Private = 48.2, Church = 60.3)
> n      <- 5
> grand_mean <- mean(means)
> grand_mean
[1] 60.1
> SS_trt <- n * sum((means - grand_mean)^2)
> SS_trt
[1] 876.9
> SS_E <- sum((n - 1) * vars)
> SS_E
[1] 725.2
> df_trt <- 2
> df_E   <- 12
> MS_trt <- SS_trt / df_trt
> MS_E   <- SS_E   / df_E
> F_stat <- MS_trt / MS_E
> F_stat
[1] 7.255102
> p_val  <- 1 - pf(F_stat, df_trt, df_E)
> p_val
[1] 0.00860219
> # 1(c): error SD
> sigma_hat <- sqrt(MS_E)
> sigma_hat
[1] 7.773888
> # 1(d): pairwise differences, Tukey HSD and Scheffé
> SE_diff <- sqrt(2 * MS_E / n)
> SE_diff
[1] 4.916638
> diffs <- c(
+   Priv_Pub = means["Private"] - means["Public"],
+   Ch_Pub   = means["Church"]   - means["Public"],
+   Ch_Priv  = means["Church"]   - means["Private"]
+ )
> diffs
Priv_Pub.Private    Ch_Pub.Church    Ch_Priv.Church
```

```

      17.4          2.7          -14.7
> qcrit <- qtkey(0.95, nmeans = 3, df = df_E)
> qcrit
[1] 3.772929
> HSD <- qcrit * sqrt(MS_E / n)
> HSD
[1] 13.11692
> Fcrit <- qf(0.95, df1 = df_trt, df2 = df_E)
> Fcrit
[1] 3.885294
> k_S <- (3 - 1) * Fcrit
> k_S
[1] 7.770588
> T2 <- (diffs^2) / (SE_diff^2)
> T2
Priv.Pub.Private    Ch.Pub.Church    Ch.Priv.Church
      12.524545         0.301572         8.939189
> #####
> # Problem 2
> #####
>
> # 2(a): One-way ANOVA from summary stats
> means_P2 <- c(G1 = 72.3, G2 = 79.5, G3 = 65.8)
> n_P2 <- 6
> a_P2 <- 3
> N_P2 <- a_P2 * n_P2
> overall_var <- 84.03
> MS_E_P2 <- 57.84
> grand_mean_P2 <- mean(means_P2)
> grand_mean_P2
[1] 72.53333
> SS_T_P2 <- overall_var * (N_P2 - 1)
> SS_T_P2
[1] 1428.51
> SS_trt_P2 <- n_P2 * sum((means_P2 - grand_mean_P2)^2)
> SS_trt_P2
[1] 563.56
> SS_E_P2_alt <- SS_T_P2 - SS_trt_P2
> SS_E_P2_alt
[1] 864.95
> df_trt_P2 <- a_P2 - 1
> df_E_P2 <- N_P2 - a_P2
> SSE_from_MS <- MS_E_P2 * df_E_P2
> SSE_from_MS

```

```

[1] 867.6
> SS_E_P2 <- SSE_from_MS
> SS_E_P2
[1] 867.6
> SS_T_P2_consistent <- SS_trt_P2 + SS_E_P2
> SS_T_P2_consistent
[1] 1431.16
> MS_trt_P2 <- SS_trt_P2 / df_trt_P2
> MS_trt_P2
[1] 281.78
> F_stat_P2 <- MS_trt_P2 / MS_E_P2
> F_stat_P2
[1] 4.871715
> p_val_P2 <- 1 - pf(F_stat_P2, df_trt_P2, df_E_P2)
> p_val_P2
[1] 0.023428
> # 2(c): pairwise differences vs control (group 2)
> SE_diff_P2 <- sqrt(2 * MS_E_P2 / n_P2)
> SE_diff_P2
[1] 4.3909
> diffs_vs_control <- c(
+   G1_minus_G2 = means_P2["G1"] - means_P2["G2"],
+   G3_minus_G2 = means_P2["G3"] - means_P2["G2"]
+ )
> diffs_vs_control
G1_minus_G2.G1 G3_minus_G2.G3
          -7.2          -13.7
> t_stats_vs_control <- diffs_vs_control / SE_diff_P2
> t_stats_vs_control
G1_minus_G2.G1 G3_minus_G2.G3
    -1.639755    -3.120090
> # 2(d): contrast = average of groups 1 and 3 vs control
> C_hat <- 0.5 * means_P2["G1"] + 0.5 * means_P2["G3"] - means_P2["G2"]
> C_hat
      G1
-10.45
> SE_C <- sqrt(MS_E_P2 * 1.5 / n_P2)
> SE_C
[1] 3.802631
> t_C <- C_hat / SE_C
> t_C
      G1
-2.748098
> alpha      <- 0.05

```

```

> K <- 10
> alpha_star <- alpha / K
> alpha_star
[1] 0.005
> t_crit_Bonf <- qt(1 - alpha_star / 2, df = df_E_P2)
> t_crit_Bonf
[1] 3.286039
> #####
> # Problem 3
> #####
>
> # 3(a): One-way ANOVA on raw gene expression
> dat <- read.csv("Stats 490/Exam1/Mid1Prob3.csv")
> str(dat)
'data.frame': 30 obs. of 4 variables:
 $ X : int 1 2 3 4 5 6 7 8 9 10 ...
 $ Project : int 1 2 3 4 5 6 7 8 9 10 ...
 $ Treatment : chr "MP Only" "MP Only" "MP Only" "MP Only" ...
 $ Gene.Expression: num 334.5 31.6 701 41.2 61.2 ...
> head(dat)
  X Project Treatment Gene.Expression
1 1      1    MP Only          334.5
2 2      2    MP Only           31.6
3 3      3    MP Only          701.0
4 4      4    MP Only           41.2
5 5      5    MP Only           61.2
6 6      6    MP Only           69.6
> dat$Treatment <- factor(dat$Treatment)
> library(dplyr)
> group_stats <- dat %>%
+   group_by(Treatment) %>%
+   summarise(
+     n = n(),
+     mean = mean(Gene.Expression),
+     sd = sd(Gene.Expression)
+   )
> group_stats
# A tibble: 3 × 4
  Treatment      n mean  sd
  <fct>      <int> <dbl> <dbl>
1 MP + HDMTX    10  479.  371.
2 MP + LDMTX    10  165.  126.
3 MP Only       10  238.  308.
> fit_raw <- aov(Gene.Expression ~ Treatment, data = dat)

```

```

> summary(fit_raw)
      Df Sum Sq Mean Sq F value Pr(>F)
Treatment    2  538442   269221   3.249 0.0544 .
Residuals   27 2236974    82851
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
> anova_raw <- summary(fit_raw)[[1]]
> anova_raw
      Df Sum Sq Mean Sq F value Pr(>F)
Treatment    2  538442   269221   3.2495 0.05439 .
Residuals   27 2236974    82851
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
> # 3(b): normality check for raw residuals
> res_raw <- residuals(fit_raw)
> par(mfrow = c(1, 2))
> qqnorm(res_raw, main = "QQ Plot - Raw Data Residuals")
> qqline(res_raw)
> hist(res_raw, main = "Histogram - Raw Data Residuals",
+       xlab = "Residuals")
> par(mfrow = c(1, 1))
> shapiro_raw <- shapiro.test(res_raw)
> shapiro_raw

```

#### Shapiro-Wilk normality test

```

data: res_raw
W = 0.91654, p-value = 0.02183

> # 3(c): log transform and ANOVA
> dat$logGE <- log(dat$Gene.Expression)
> group_stats_log <- dat %>%
+   group_by(Treatment) %>%
+   summarise(
+     n      = n(),
+     mean   = mean(logGE),
+     sd     = sd(logGE)
+   )
> group_stats_log
# A tibble: 3 × 4
  Treatment      n  mean    sd
  <fct>      <int> <dbl> <dbl>
1 MP + HDMTX    10  5.73  1.13
2 MP + LDMTX    10  4.74  1.00

```

```

3 MP Only      10  4.79  1.18
> fit_log <- aov(logGE ~ Treatment, data = dat)
> summary(fit_log)
      Df Sum Sq Mean Sq F value Pr(>F)
Treatment    2   6.30    3.148   2.571 0.0951 .
Residuals   27  33.07    1.225
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
> anova_log <- summary(fit_log)[[1]]
> anova_log
      Df Sum Sq Mean Sq F value  Pr(>F)
Treatment    2   6.297    3.1485   2.5707 0.09507 .
Residuals   27  33.069    1.2248
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
> # 3(d): residual diagnostics for log model
> res_log <- residuals(fit_log)
> fitted_log <- fitted(fit_log)
> par(mfrow = c(1, 2))
> plot(fitted_log, res_log,
+      main = "Residuals vs Fitted (log model)",
+      xlab = "Fitted values", ylab = "Residuals")
> abline(h = 0, col = "red")
> qqnorm(res_log, main = "QQ Plot - log model residuals")
> qqline(res_log, col = "red")
> par(mfrow = c(1, 1))
> shapiro_log <- shapiro.test(res_log)
> shapiro_log

```

Shapiro-Wilk normality test

```

data:  res_log
W = 0.95648, p-value = 0.2511

```